



Fig. 5: senseFly eBee SQ

Technical Specifications of senseFly eBee SQ		Table 4
Specification Type	Value	
Flight Type	Fully autonomous navigation including take-off and landing and automatic control of the on-board camera	
Wingspan	110cm	
Weight (incl. supplied camera and battery)	1.1kg	
Cruise Speed	40-110km/h	
Cameras Supplied	Parrot Sequoia	
Max. Flight Time: Standard: 55 minutes Max. Flight Range: Standard: 41km Nominal Coverage at 120m: 200ha Nominal Coverage at 2,000m: 30km ² Landing: Automatic, linear with ~5m (16.4ft) accuracy		

Technical Specifications of Delair UX 11 AG		Table 5
Endurance	Up to 52 minutes	
Weight	1.6kg	
Deployment Time	5 minutes	
Cruise Speed	54km/h (33.5mph)	
Take-off/Landing	Hand-launched (angle: 30 degrees)/Belly landing (angle: 30 degrees)	
Flying Range	47km	
3G/4G	Unlimited (within network coverage)	
Optional Radio Link	FCC upto 10km; CE upto 5km	
Image Resolution	GSD down to 5cm	
Sensor Type	Global shutter, distortion-free	



Fig. 6: Delair UX11 AG

Artificial Intelligence

A drone technology in development involves machine learning. Improving artificial intelligence (AI) in drones is important to make them more useful to smaller farmers in developing nations. Current drone technologies are more effective in monitoring well-known crops like corn that are planted in large monocultural field patterns. Drone monitoring programs, as they stand, have a hard time recognising areas with increased crop diversity, less well-known produce, and grains which look similar throughout their growth stages, and so are less effective in monitoring such crop growth and their health.

Commercial drones

Companies such as DJI have developed customised models for crop protection against insects or weeds by carrying out the integration of specific sensors such as multispectral, high-precision RTK GPS, and high-resolution cameras. Parrot Bluegrass Fields is a complete drone solution having drone + multispectral sensor + processing software, ready to fly and suitable for the entire crop analysis workflow. Parrot Bluegrass Fields provides farmers with information to maximize their yield and improve the quality of their crops.

There is always a trade-off between endurance (maximum time of flight) and payload in drones. Thus, if endurance increases (by adding more powerful batteries or a higher capacity fuel tank), the payload decreases due to the increment in weight. Table 1 highlights some of the main drone types used in agriculture and their characteristics.

Sensors in agricultural drones

The reduction in weight and consumption of newer equipment and measurement instruments has made it possible to improve the drone sensors and therefore to expand their field of application. Table 2 highlights various sensors integrated with agricultural drones classified into RGB (visible spectrum) sensors, multi-spectral and hyperspectral cameras. In addition, drones also integrate laser, thermal, and multi-scanner sensors.